

Bibee 2026 Integrals WITH SOLUTIONS

Integral selection committee

May 2026

1 Kahoot

1. Series-ly???

$$\int_0^\pi \sum_{k=1}^{2026} \frac{1 + \cos(2kx)}{2} dx$$

Solve by switching sum and integral. $\forall k \in \mathbb{Z} : \int_0^\pi \cos(2kx) dx = 0$ and

$$\int_0^\pi \frac{1}{2} dx = \frac{\pi}{2}. \text{ We get } \boxed{1013\pi} \text{ as our solution.}$$

2. I know this one!

$$\int_{-\infty}^0 x^2 e^{x^3} dx$$

Substitute $u = x^3$. The solution is $\boxed{\frac{1}{3}}$.

3. Watch out!

$$\int_{-\infty}^{\infty} e^{-z^2} dx$$

This is NOT a Gaussian integral because we are integrating with respect to z instead of x . The right solution is $\boxed{\infty}$.

4. Watch out! (Part 2)

$$\int_{-\infty}^{\infty} \lfloor e^{-\lceil x^2 \rceil} \rfloor dx$$
$$= \boxed{0}$$

5. Trig-Trag-Troe.

$$\int \frac{dx}{\sec(x) + \tan(x) \sin(x)}$$

We multiply top and bottom by $\cos(x)$, substitute $u = \sin(x)$ and get

$$\boxed{\tan^{-1}(\sin(x))}.$$

6. Determinated.

$$\int_0^{\frac{1}{3}} 3 \det \begin{pmatrix} 1 & e^{-x} & e^x \\ e^x & 1 & e^{-x} \\ e^{-x} & e^x & 1 \end{pmatrix} dx$$

Sarrus gives us the determinant: $e^{-3x} + e^{3x} - 2$. We integrate and get

$$\boxed{e - \frac{1}{e} - 2}.$$

7. **Funny?**

$$\begin{aligned} & \int_6^7 (6x^2 - 7x^2)^{6-7} dx \\ &= - \int_6^7 \frac{1}{x^2} dx = -\frac{1}{6} + \frac{1}{7} = \boxed{-\frac{1}{42}} \end{aligned}$$

8. **Yes, you have to actually do this**

$$\begin{aligned} & \int_0^{24} \lfloor \sin x \rfloor dx \\ & \text{Split up the integral at } \pi, 2\pi, 3\pi, 4\pi, 5\pi, 6\pi, 7\pi \text{ to get } 0 - \pi + 0 - \pi + \\ & 0 - \pi + 0 - (24 - 7\pi) = \boxed{4\pi - 24} \end{aligned}$$

9. **Beep beep**

$$\begin{aligned} & \int_0^\pi \frac{1}{1 + \tan x} + \frac{1}{1 + \cot x} dx \\ &= \int_0^\pi \frac{\cos x}{\sin x + \cos x} + \frac{\sin x}{\sin x + \cos x} dx = \boxed{\pi} \end{aligned}$$

10. **Can you FIX it?**

$$\begin{aligned} & \int_0^1 [\cos(\arctan(x)) + \sin(\arctan(x))]^2 + [\cos(\arctan(x)) - \sin(\arctan(x))]^2 dx \\ &= \boxed{2}. \text{ Proof by obviousness.} \end{aligned}$$

11. **How fast am I, going to infinity?**

$$\begin{aligned} & \int_0^x \cos(\cos(\cos(\cos(\cos(\dots(t)\dots)))))) dt \\ & \text{Since infinite cosines are leaving us with a constant (tryhards use Banach's} \\ & \text{Fix Point Theorem on this), it is } \boxed{\Theta(x)}. \end{aligned}$$

12. **Dividing Polynomials**

$$\begin{aligned} & \int_0^1 \frac{x^3 + 3x^2 + 3x}{x + 1} dx \\ &= \int_0^1 \frac{(x + 1)^3 - 1}{x + 1} dx = \int_1^2 \frac{y^3 - 1}{y} dy = \boxed{\frac{7}{3} - \ln(2)} \end{aligned}$$

13. **What's our favorite animal?**

$$\begin{aligned} & \int_0^2 \int_0^2 \int_0^2 \int_0^2 \int_0^2 beeee db de de de de \\ &= \int_0^2 \int_0^2 \int_0^2 \int_0^2 2e^4 de de de de = \int_0^2 \int_0^2 \int_0^2 \frac{64}{5} de de de = \frac{64 \cdot 2^3}{5} = \\ & \boxed{\frac{512}{5}}. \end{aligned}$$

14. **The world's smallest violin!**

$$\int_0^{0.2026} \frac{1}{2\sqrt{|\text{♩}-0.0001|}} d\text{♩}$$

$$\text{This equals } \operatorname{sgn}(\text{♩})\sqrt{|\text{♩}|}\Big|_{-0.0001}^{0.2025} = 0.45 + 0.01 = \boxed{0.46}.$$

15. **Another 2026!**

$$\begin{aligned} & \int_0^{\sqrt{2026}} \frac{(2x)^2}{\sqrt{2026-x^2}} dx \\ &= 4 \int_0^{\pi/2} 2026 \sin^2(u) du = 4 \cdot 2026 \left(\frac{u}{2} - \frac{\sin(2u)}{4} \right) \Big|_0^{\pi/2} = \boxed{2026\pi} \end{aligned}$$

substituting $x = \sqrt{2026} \sin(u)$ and using your favorite way of integrating $\sin^2$.

16. **Weeeeeeeeeeeeeeeeeee!**

$$\begin{aligned} & \int_0^1 e^{e^{e^{e^{e^x}}}} e^{e^{e^{e^x}}} e^{e^{e^x}} e^{e^x} e^x dx \\ &= \left[e^{e^{e^{e^{e^x}}}} \right]_{x=0}^1 = \boxed{e^{e^{e^{e^e}}} - e^{e^{e^e}}}. \end{aligned}$$

17. **Trippy greeting from the MIT :O**

$$\int \int \int \int \int_0^1 x dx \int_0^1 x dx \int_0^1 x dx \int_0^1 x dx \int_0^1 x dx = \int_{\frac{55}{128}}^{\frac{1}{128}} x dx = \boxed{-\frac{189}{2048}}.$$

Remark: This is stolen from the MIT Integration Bee 2026.

18. **A radical product.**

$$\int_0^1 \prod_{n=1}^{\infty} \sqrt[n^2]{1+x} dx$$

Use that $\prod_{n=1}^{\infty} (1+x)^{\frac{1}{n^2}} = (1+x)^{\sum_{n=1}^{\infty} \frac{1}{n^2}} = (1+x)^{\frac{\pi^2}{6}}$. The solution

$$\boxed{\frac{2^{\frac{\pi^2}{6}+1}-1}{\frac{\pi^2}{6}+1}} \text{ follows directly.}$$

19. **Is it even integrable?**

$$\begin{aligned} & \int x^{\frac{\ln \sin x}{\ln x}} dx \\ &= \int \sin x dx = \boxed{-\cos x}. \end{aligned}$$

20. **A famous one**

$$\int_1^{\infty} \frac{1}{\lceil x \rceil^2} dx$$

$$= \boxed{\frac{\pi^2}{6} - 1}$$

21. **Sneaky Euler.**

$$\int_0^{\frac{\pi}{2}} e^{ix} \cos(x) dx$$

We use $\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$, simplify via algebraic manipulation and Euler's

identity. Solution: $\boxed{\frac{\pi}{4} + \frac{i}{2}}$.

22. **Back to the roots!**

$$\int_0^5 \sqrt{x \sqrt{x \sqrt{x \dots}}} dx$$

By the identity: $x^{1/2+1/4+\dots} = x^1$ we get above is $\int_0^5 x = 12.5$

23. **Log salad.**

$$\int_{e^2}^{e^4} \frac{\ln 2x}{x \ln x} dx$$

$$= \int_{e^2}^{e^4} \frac{\ln x + \ln 2}{x \ln x} dx = \ln x + \ln 2 \ln \ln x \Big|_{e^2}^{e^4} = \boxed{2 + \ln^2 2}$$

24. **OG change of basis**

$$\int_0^1 5^{\log_2 x} dx$$

$$= \int_0^1 x^{\frac{1}{\log_5 2}} dx = \frac{1}{1 + \frac{1}{\log_5 2}} = \boxed{\log_{10} 2}$$

25. **Welcome to Bonn Integration Bee**

$$\int_0^{\infty} 2026^{-x} dx$$

$$= \frac{1}{\ln 2026}$$

2 Tiebreakers

1. **QuadWave!**

$$\int \sin^2(x) \cos^2(x) dx$$

$$= \frac{1}{4} \int \sin^2(2x) dx = \frac{1}{8} \int \sin^2(u) du = \boxed{\frac{x}{8} - \frac{\sin(2x) \cos(2x)}{16}}$$

Mostly irrelevant comment by Leandro: This is equivalent to $\frac{x}{8} - \frac{\sin(4x)}{32}$.

2. **I hope you have good eyes.**

$$\int_0^1 2026202620262026202620262026^x dx =$$

$$\boxed{\frac{(202620262026202620262026\textcolor{red}{2025})}{\ln(2026202620262026202620262026)}}$$

3. **Geometry... without triangles!?**

$$\int \frac{(1+x+x^2+\dots+x^{2024})(1-x^3)}{1+x+x^2} dx$$

The integrand simplifies to $1-x^{2025}$, so the answer is $\boxed{x - \frac{1}{2026} x^{2026}}$

4. **Is this what you think?**

$$\int_0^3 4\lfloor x \rfloor^{\lceil x \rceil^{\lfloor x \rfloor}} + \lceil x \rceil^{\lfloor x \rfloor} - \lfloor x \rfloor^{\lfloor x \rfloor} dx$$

$$= \boxed{2080}$$

This is not = 2026 because we f****d up a little.

5. **Neper's Root**

$$\int \sqrt{e^x - 1} dx$$

Substitute $t = \sqrt{e^x - 1}$ to get $\int \frac{2t^2}{1+t^2} dt$, then standard techniques lead to

$$\boxed{2\sqrt{e^x - 1} - 2 \arctan \sqrt{e^x - 1}}.$$

6. **See you on the 3rd floor.**

$$\int_0^3 2(\lfloor x \rfloor^{\lceil x \rceil^{\lfloor x \rfloor}} + 250\lfloor \sqrt{x} \rfloor) dx$$

We evaluate piecewise and get $2(0+1+512)+500(2) = \boxed{2026}$

7. **There is always one trig integral more.**

$$\int \frac{dx}{9\cos^2(x) + 4\sin^2(x)}$$

Use the substitution $t = \tan x$ to get $\int \frac{dt}{9+4t^2}$, so the solution is

$$\boxed{\frac{1}{6} \arctan\left(\frac{2 \tan(x)}{3}\right)}$$

8. **Double root.**

$$\begin{aligned} & \int_0^1 \sqrt{x} e^{\sqrt{x}} dx \\ &= 2 \int_0^1 u^2 e^u du = 2(u^2 - 2u + 2)e^u \Big|_0^1 = 2e - 4 \end{aligned}$$

9. **From floor to ceiling.**

$$\begin{aligned} & \int_0^5 \left(\left\lfloor \frac{x}{2} \right\rfloor - \left\lceil \frac{x}{3} \right\rceil \right) dx \\ &= \int_0^2 dx + \int_2^3 2 dx + \int_3^4 dx + \int_4^5 2 dx = \boxed{7} \end{aligned}$$

10. **Where does π come from?**

$$\begin{aligned} & \int_{1/e}^e \arctan x \frac{(\ln x)^{2026}}{x} dx \\ & \text{Call the integral } I, \text{ use the substitution } t = 1/x \text{ to see } I = \int_{1/e}^e \arctan \frac{1}{x} \frac{(\ln x)^{2026}}{x} dx, \\ & \text{then sum up using } \arctan x + \arctan \frac{1}{x} = \frac{\pi}{2} \text{ so that } 2I = \frac{\pi}{2} \int_{1/e}^e \frac{(\ln x)^{2026}}{x} dx = \\ & \frac{\pi}{2} \left[\frac{(\ln x)^{2027}}{2027} \right]_{1/e}^e = \frac{\pi}{2027}. \text{ Finally } \boxed{I = \frac{\pi}{4054}} \end{aligned}$$

11. **Happy New Year!?**

$$\begin{aligned} & \lim_{R \rightarrow \infty} \int_{-R}^R \prod_{j=0}^{2026} \frac{1}{j+1} x^{j^2+j+1} dx \\ & j^2 + j + 1 = j(j+1) + 1 \text{ is odd. Hence, the integral collapses to } \boxed{0}. \end{aligned}$$

12. **The last integral of the evening is really shit.**

$$\begin{aligned} & \int_0^1 x^{13} + 67x + x^9 + 13x^{12} + 3x^6 + 2(x+1)^{-3} dx \\ &= \boxed{\frac{5019}{140}} \end{aligned}$$

3 QF

1. **Root of all Problems.**

$$\int_1^2 \frac{x^5}{\sqrt{x^3-1}} dx$$

$$= \frac{1}{3} \int_1^8 \frac{u}{\sqrt{u-1}} du = \frac{1}{3} \int_1^8 \sqrt{u-1} + \frac{1}{\sqrt{u-1}} du = \boxed{\frac{20\sqrt{7}}{9}}$$

2. **Antagonist views**

$$\int \frac{4x^2 + 6x + 1}{x^3 + 2x^2 + x + 2} dx$$

$$= \int \frac{1}{x+2} + \frac{3x}{x^2+1} dx = \boxed{\ln|x+2| + \frac{3}{2} \ln(x^2+1)}$$

3. **Walking in circles.**

$$\int_0^{2\pi} \prod_{n=0}^{2026} (n^2 + 1) \cos\left(\frac{x}{3} \prod_{n=0}^{2026} (n^2 + 1)\right) dx$$

Write $A = \prod_{n=0}^{2026} (n^2 + 1)$. The integral is $\int_0^{2\pi} A \cos\left(\frac{x}{3} A\right) dx$ or $3 \int_0^{2\pi A/3} \cos x dx$. It can be computed that A is an integer 2 modulo 3. By periodicity, the

$$\text{answer is } 3(\sin(\frac{4\pi}{3}) - \sin(0)) = \boxed{-\frac{3\sqrt{3}}{2}}.$$

4. **Fraction Action!?**

$$\int_0^\infty \frac{\{x\}}{([x]+1)^2} dx$$

Write the integral as $\sum_{n=0}^\infty \int_n^{n+1} \frac{x-n}{(n+1)^2} dx = \sum_{n=0}^\infty \frac{1}{(n+1)^2} \int_0^1 u du$

$$= \frac{1}{2} \sum_{n=0}^\infty \frac{1}{(n+1)^2} = \frac{1}{2} \cdot \frac{\pi^2}{6} = \boxed{\frac{\pi^2}{12}}$$

5. **Unreasonably rational!**

$$\int \frac{2x^2 + 3x + 1}{x^3 + x^2 + x} dx$$

Solve with partial fractions: $\frac{2x^2+3x+1}{x^3+x^2+x} = \frac{1}{x} + \frac{x+2}{x^2+x+1}$.

Note that $x+2 = \frac{1}{2}(2x+1) + \frac{3}{2}$. To evaluate $\int \frac{1}{x^2+x+1} dx$, we complete

the square and calculate $\int \frac{1}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} dx = \frac{2}{\sqrt{3}} \arctan\left(\frac{2x+1}{\sqrt{3}}\right)$.

The solution is given by $\boxed{\ln|x| + \frac{1}{2} \ln(x^2+x+1) + \sqrt{3} \arctan\left(\frac{2x+1}{\sqrt{3}}\right)}$

6. **Digitization!**

Let $s(n)$ denote the **digit sum of n**. / Sei $s(n)$ die **Quersumme von n**.

$$\int_0^1 s(\lfloor 100x \rfloor) dx$$

Because of linearity, the integral equals $\sum_{k=0}^{99} \int_{\frac{k}{100}}^{\frac{k+1}{100}} s(k) dx = \frac{1}{100} \sum_{k=0}^{99} s(k)$.

Let $k = 10a + b$, $a, b \in \{0, \dots, 9\}$ such that $s(n) = a + b$.

It follows: $\sum_{k=0}^{99} s(n) = \sum_{a=0}^9 \sum_{b=0}^9 (a + b) = 10 \sum_{a=0}^9 a + 10 \sum_{b=0}^9 b = 900$.

So the solution is $\frac{900}{100} = \boxed{9}$.

7. Dotsy and messy

$$\int_0^1 \sqrt{x^2 + x + \sqrt{x^2 + x + \sqrt{\dots}}} dx$$

Calling the integrand y we have by recursion $y = \sqrt{x^2 + x + y}$, so that $y^2 - y = x^2 + x$, hence $y - \frac{1}{2} = \pm(x + \frac{1}{2})$. By positivity $y = x + 1$ so we

are left with $\int_0^1 (x + 1) dx$ and the answer is $\boxed{\frac{3}{2}}$

Comment: For this kind of integrals one should of course be careful of convergence, I think we can omit and just use everything as if the convergence is given.

8. Some ancient Greek would be proud for sure...

$$\int_0^1 \arctan\left(\frac{\sqrt{1-x^2}}{x}\right) dx$$

substitute $x = \cos(y)$ and get $\boxed{1}$ or do some really shitty workaround using Thales' theorem like Leandro xD

4 SF

1. **If you know, you know.**

$$\begin{aligned} & \int_0^{\pi/4} \frac{\sqrt{\tan(x)}}{\cos^4(x)} dx \\ &= \int_0^1 \frac{\sqrt{u}}{\cos^2(\arctan(u))} du = \int u^{5/2} + u^{1/2} du = \left[\frac{2}{7} u^{7/2} + \frac{2}{3} u^{3/2} \right]_0^1 = \boxed{\frac{2}{7} + \frac{2}{3}} \\ & \text{substituting } u = \tan x \text{ and using } \cos(\arctan(u)) = \frac{1}{\sqrt{u^2+1}}. \end{aligned}$$

2. **Polynomial Chaos!**

$$\begin{aligned} & \int_{\ln(\frac{e^2}{\sqrt{5}})}^{\ln(e^2\sqrt{5})} (x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 30)(\cos(x) \cos(2) + \sin(x) \sin(2)) dx \\ &= \int_{\ln(\frac{e^2}{\sqrt{5}})}^{\ln(e^2\sqrt{5})} (x-2)^5 \cos(x-2) dx + 2 \int_{\ln(\frac{e^2}{\sqrt{5}})}^{\ln(e^2\sqrt{5})} \cos(x-2) dx = \boxed{4 \sin(\ln(\sqrt{5}))}. \end{aligned}$$

The first is just = $\boxed{0}$ by symmetry around $x = 2$.

3. **Delicious!!**

$$\begin{aligned} & \lim_{n \rightarrow \infty} \int_1^9 \sum_{k=1}^n \frac{1}{\sqrt{n^2 x + k}} dx \\ & \text{Write } a_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2 x + k}}. \text{ First, we exchange limit and integral and} \\ & \text{compute } \lim_{n \rightarrow \infty} a_n = \frac{1}{\sqrt{x}}. \text{ This follows from } a_n \geq n \cdot \frac{1}{\sqrt{n^2 x + n}} = \frac{1}{\sqrt{x + \frac{1}{n}}} \\ & \text{and } a_n \leq n \cdot \frac{1}{\sqrt{n^2 x}} = \frac{1}{\sqrt{x}} \text{ by applying the Sandwich-lemma. Hence the} \\ & \text{integral is } \int_1^9 \frac{1}{\sqrt{x}} dx = [2\sqrt{x}]_1^9 = 6 - 2 = \boxed{4}. \end{aligned}$$

4. $\pi/3$ and $\pi/4$ would be worse boundaries

$$\begin{aligned} & \int_0^{\pi/3} \tan^2(x) x dx \\ &= (\tan x - x)x - \int_0^{\pi/3} \tan x - x dx = \left[x \tan x + \ln(\cos(x)) - \frac{x^2}{2} \right]_0^{\pi/3} = \\ & \boxed{\frac{\pi}{\sqrt{3}} - \ln 2 - \frac{\pi^2}{18}}, \text{ where } \int \tan^2(x) dx = \int \tan^2(x) + 1 - 1 dx = \tan x - x. \end{aligned}$$

5 3rd Place

1. **This Partialbruchzerlegung is nasty.**

$$\begin{aligned} & \int_{\sqrt{2}}^{\sqrt{3}} \frac{x^3(x^2+3)}{(x^4-1)(x^2+1)} dx \\ &= \int_{\sqrt{2}}^{\sqrt{3}} \frac{x^5+3x^3}{x^6+x^4-x^2-1} dx = \frac{1}{2} \int_2^3 \frac{u^2+3u}{u^3+u^2-u-1} du = \frac{1}{2} \int_2^3 \frac{1}{(u+1)^2} + \\ & \frac{1}{u-1} du = \boxed{\frac{1}{24} + \frac{\ln(2)}{2}} \end{aligned}$$

The trick is to see, that you can substitute $u = x^2$, then the partial fractions calculation becomes easy.

2. **Modest but beautiful.**

$$\begin{aligned} & \int \frac{\sin(x)}{\sin(x)+1} dx \\ &= \int 1 - \frac{1}{\sin(x)+1} dx = x - \int \frac{1-\sin(x)}{\cos^2(x)} dx = \boxed{x - \tan(x) + \sec(x)} \end{aligned}$$

3. **Tricky Trigs.**

$$\int_0^{\frac{\pi}{2}} (\sin(x))^{2\cos(x)-1} \cos^2(x) - \sin(x)^{2\cos(x)+1} \ln(\sin(x)) dx$$

Basic algebraic manipulation gives us

$$\int_0^{\pi} \sin(x)^{\cos(x)} [\sin(x)^{\cos(x)} \left(\frac{\cos^2(x)}{\sin(x)} - \sin(x) \ln(\sin(x)) \right)] dx,$$

substitute $u = \sin(x)^{\cos(x)}$ since $u' = \sin(x)^{\cos(x)} \frac{\cos^2(x)}{\sin(x)} - \sin(x) \ln(\sin(x))$.

We evaluate and get $\boxed{\frac{1}{2}}$ as our solution.

This is exactly the derivative of $\sin^{2\cos}$.

4. **Quotients**

$$\begin{aligned} & \int_1^{\infty} \frac{e^{-x}}{x} + \frac{e^{-x}}{x^2} dx \\ &= \int_1^{\infty} \frac{e^{-x}}{x} dx + \left(-\frac{e^{-x}}{x} \Big|_1^{\infty} - \int_1^{\infty} \frac{e^{-x}}{x} dx \right) = -\frac{e^{-x}}{x} \Big|_1^{\infty} = \boxed{e^{-1}} \end{aligned}$$

You also could just invert the quotient rule immediately.

6 Final

1. Roots bloody roots!

$$\int_0^\infty \frac{\ln x}{(1+x\sqrt{2})\sqrt{2}} dx$$

Just use $t = x^{-1}$ and symmetry to get $\boxed{0}$, but it's cool to see :)

2. Can you spot the ODE?

$$\begin{aligned} & \int_{-\infty}^0 \frac{(x + \sqrt{1+x^2})^{2026}}{\sqrt{1+x^2}} dx \\ &= \int_0^1 u^{2025} du = \boxed{\frac{1}{2026}}, \text{ substituting } u = x + \sqrt{1+x^2} \text{ and using, that} \\ & \frac{d}{dx} x + \sqrt{1+x^2} = \frac{x + \sqrt{1+x^2}}{\sqrt{1+x^2}}. \end{aligned}$$

3. Pure Purgatory.

$$\lim_{n \rightarrow \infty} \sqrt{\pi n} \int_0^1 \frac{x^{2n}}{\sqrt{1-x^2}} dx$$

We substitute $x = \sin(t)$ and get $\lim_{n \rightarrow \infty} \sqrt{\pi n} \int_0^{\frac{\pi}{2}} \sin^{2n}(x) dx$. This can be

solved with Wallis' Product and we get $\lim_{n \rightarrow \infty} \sqrt{\pi n} \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!}$. We either

know by heart - or show by using the Stirling formula - that $\frac{(2n-1)!!}{(2n)!!}$ is

asymptotically equivalent to $\frac{1}{\sqrt{\pi n}}$ and thus, the integral resolves to $\boxed{\frac{\pi}{2}}$.

4. Not spherical, that's hyperbolic!

$$\begin{aligned} & \int \frac{\sqrt{9 \cosh^2(x) \sinh^2(x) - 3 \sinh^2(x) + 1}}{\cosh(x) \sinh(x)} dx \\ &= \int \frac{\sqrt{9(\sinh^2(x) + 1) \sinh^2(x) - 3 \sinh^2(x) + 1}}{(\sinh^2(x) + 1) \sinh(x)} \cosh x dx \\ &= \int \frac{\sqrt{9u^4 + 6u^2 + 1}}{u^3 + u} du = \int \frac{3u^2 + 1}{u^3 + u} du = \ln(u^3 + u) = \boxed{\ln(\sinh(x) \cosh^2(x))} \\ & \text{substituting } u = \sinh(x) \text{ and using } \cosh^2 = \sinh^2 + 1. \end{aligned}$$

5. Dancing constants

$$\int_0^{\frac{\pi}{2}} \frac{e^{|x - \frac{\pi}{4}|}}{1 + \tan^\varphi x} dx.$$

Call the integral I , then by using $t = \frac{\pi}{2} - x$ one finds $I = \int_0^{\frac{\pi}{2}} \frac{e^{|x - \frac{\pi}{4}|}}{1 + \tan^\varphi x} \tan^\varphi x dx$

so that $2I = \int_0^{\frac{\pi}{2}} e^{|x - \frac{\pi}{4}|} dx$ then split $\int_0^{\frac{\pi}{4}} + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ and conclude $\boxed{e^{\frac{\pi}{4}} - 1}$.

7 Bonus for those who want real stress in their life

1. A Phi-nomenal Integral.

$$\int_0^\infty \frac{dx}{(1+x^\phi)^\phi}$$

Substitute $u = x^\phi \rightarrow \frac{1}{\phi} \int_0^\infty \frac{u^{\frac{1}{\phi}-1}}{(1+u)^\phi} du$. This resolves into a β -function.

We get $\frac{\Gamma(\frac{1}{\phi})}{\phi\Gamma(\phi)}$. Since $\phi\Gamma(\phi) \stackrel{\text{Gamma identity}}{=} \Gamma(\phi+1) \stackrel{\text{Golden ratio property}}{=} \Gamma(\frac{1}{\phi})$, the solution is $\boxed{1}$.